



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

April 23, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 12574
(ELEVENTH REVISION)

EXPIRATION DATE: 2029-03-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Weldship LLC
Bethlehem, PA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the transportation in commerce of certain compressed gases in manifolded and framed seamless steel cylinders. The cylinders are retested by acoustic emission (AE) and ultrasonic examination (UE) described in paragraph 7. below in place of the internal visual inspection and the hydrostatic retest required in § 180.205. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 180.209(a), § 180.205(c), (f), (g), and (i), and § 173.302a(b)(2), (3), (4), and (5) in that AE and UE are performed in place of the hydrostatic test and the internal visual inspection, and § 180.213 in that the retest month & year are stamped on the dataplate and affixed to the rear bulkhead of the unit when tested as a unit.

NOTE: This does not relieve the holder of this special permit from securing an approval for retesting cylinders from the Associate Administrator for Hazardous Materials Safety.

Tracking Number: 2025024057

5. BASIS: This special permit is based on the application of Weldship LLC dated February 4, 2025, and submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Compressed gas, oxidizing, n.o.s. (up to 50% oxygen)	2.2	UN3156	N/A
Compressed gas, n.o.s.	2.2	UN1956	N/A
Helium, compressed	2.2	UN1046	N/A
Nitrogen, compressed	2.2	UN1066	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Packagings prescribed are seamless steel cylinders (flanged-end tube) of 50 cubic feet nominal water capacity each, originally manufactured and maintained in conformance with the requirements for DOT Specification 107A seamless steel tank car tanks (§ 179.500 and § 180.500). The cylinders must have been removed from the trucks and are framed, manifolded, and mounted on a trailer. The packaging must meet the following:

- (1) Each cylinder must be equipped with one shut-off valve that is rated for the test pressure of the cylinder.
- (2) A minimum of 1% of the flange stud bolts must be tested for the required strength and elongation.
- (3) Each flange gasket must be leak tight and would not be prone to cold flow for the maximum operating pressure at 149 °F (65 °C) temperature when the cylinder is filled with compressed gas.

(4) Each cylinder must be equipped with one or more pressure relief devices in accordance with § 173.301(f)(1) and (g)(1).

b. TESTING: Each cylinder (tank) must be retested and reinspected as prescribed in § 180.205 for DOT-3AAX cylinders, except that the cylinder is examined by the acoustic emission (AE) and ultrasonic examination (UE) method described below in place of the hydrostatic pressure test and internal visual inspection. A cylinder that has been exposed to fire or to excessive heat (temperatures of 1,000 °F or greater) must not be retested under the terms of this special permit.

(1) Visual Examination: Each tube must receive an external visual examination in accordance with CGA Pamphlet C-6.

(2) Acoustic Emission (AE) Equipment: The AE equipment must be in accordance with the specification described in Weldship Corporation's application on file with the Office of Hazardous Materials Safety (OHMS) and as prescribed in this special permit. Power supply, signal cable, signal processor, and couplant must meet all requirements of the American Society for Testing and Materials (ASTM), 1419-02b Standard Test Method for Examination of Seamless, Gas-Filled, Pressure Vessels Using Acoustic Emission.

(3) AE Calibration and Standardization: Calibration of AE equipment performance and test procedure must be in accordance with ASTM E 1419-02b and Weldship Corporation's test method on file with OHMS, except as specifically stated herein.

(4) AE Test Pressurization:

(i) The test pressure must equal 1.1 times the highest fill pressure experienced by any individual cylinder in the unit for the last 12 months prior to requalification. If the highest fill pressure is not available the test pressure must be 1.20 times the lowest service pressure stamped cylinder in the unit.

(ii) The data collection must begin at a pressure less than or equal to 50% of the AE test pressure. The rate of pressurization must be such that the saturation of electronic circuitry does not occur.

NOTE: AE retest per this special permit shall not be used to requalify a cylinder that had been hydrotested within the last 1 year.

(5) AE Test Evaluation and Markup:

(i) Each AE site on the cylindrical portion of each cylinder (tube) that produces 5 or more events within an 8-inch (204 mm) axial distance must be marked for the follow up UE as prescribed herein.

(ii) Each AE site on the tube ends (i.e., sections of the tube that lie outboard of the sensors), which produces five or more events that hit both sensors and that had 43 dB or greater peak amplitude at the “first hit” sensor, must be subjected to UE by using shear wave.

(6) Ultrasonic Examination (UE) Equipment (Apparatus): The UE equipment must be in accordance with the specification described in ASTM E-2223-02 “Examination of Seamless, Gas Filled, Steel Pressure Vessels, Using Angle Beam Ultrasonics”.

(i) Each search unit used must have the appropriate frequencies (1-5 MHz) and refracted angle (45-75°) for the material and geometry of the cylinder that is being examined. Other angles and frequency combinations as found appropriate may be used for flaw sizing.

(ii) Each search unit must detect and display the indication from the notch on reference ring at the maximum distance to be used during the examination.

(iii) The search unit must be comprised of a transducer mounted on a plastic wedge that is designed to have continuous acoustic coupling between the search unit and the cylinder wall.

(7) UE Standardization Ring With Reference Notches (Reference Ring):

(i) The reference ring must be fabricated from the same type of cylinder that is being examined. The reference ring must have the same diameter, minimum design wall thickness with a tolerance of +/- 10%, material, heat treatment and surface condition as the cylinder to be examined. Reference notches must be placed into both internal and external surfaces of the reference ring. Notches must be made by EDM process. One or more notches may be placed into a single reference ring. Each circumferential notch must have a depth of less than or equal to 25 percent of t_m or 0.060 in. (1.53 mm), a width of less than or equal to 0.020 in. (0.5 mm) and a length of 1 in. (25.4 mm).

(ii) A certification statement signed by a Weldship LLC Senior Review Technologist (SRT) must be available for all standard references at each site

where retesting is performed. The certification statement must include a standard reference drawing for each size and type of cylinder. A standard reference drawing must include dimensions and the locations of each simulated defect.

(8) UE Standardization Procedure: The UE equipment must be standardized for each cylinder type by using the Standardization Ring and in accordance with procedure described in ASTM E 2223-02.

(9) UE Procedure:

(i) The UE of each cylinder must be in accordance with the ultrasonic examination described in ASTM E 2223-02 except that:

(A) The extent of the examination shall be 18 inches on either side of the axial location (on the cylindrical portion) as determined through AE.

(B) The examination shall be performed within a single “V-path” if any flaw indication exceeds 20% of DAC. For indications exceeding 20% of DAC, flaw characterization and sizing shall be performed by TOFD or other suitable technique.

(ii) A copy of the most recent approved operating test procedure must be at each facility performing ultrasonic examination and be made available to a DOT representative when requested. Any change to the written procedures or in UE equipment (software or hardware), other than as supplied by the original equipment manufacturer, must be submitted to and approved by AAHMS prior to implementation.

(10) Rejection Criteria: Rejection criteria as established by fracture mechanics for the cylinder retested under this special permit is a flaw with a maximum depth less than or equal to 25% of the minimum design wall thickness or any crack with a maximum depth of 0.060 in. (1.53 mm). When a cylinder is rejected, the retester must stamp a series of X’s over the special permit or DOT specification number and marked test pressure, or stamp “CONDEMNED” on the shoulder or neck using a steel stamp, and must notify the cylinder owner, in writing, that the cylinder is rejected and may not be filled with hazardous material for transportation in commerce. Alternatively, at the direction of the owner, the retester may render the cylinder incapable of holding pressure.

c. MARKING: Each cylinder must be marked “DOT-SP 12574” in place of the DOT-107A specification mark in characters not less than 1/2 inch high. The existing DOT-107A marking may be removed or permanently covered to clarify the cylinder is

authorized for over the road usage. Each cylinder passing retests under the provisions of this special permit must be marked as prescribed in § 180.213(d). In addition, each cylinder must be marked AE/UE, in characters not less than 1/4 inch high at a location close to the retester's marking.

d. REPORT: A report must be generated for each unit listing all cylinders that were examined. The AE and UE reports must be on file at the test site, and made available to a DOT official when requested. The AE and UE reports must include the following:

- (1) Unit/Module/Batch control identification.
- (2) AE and UE equipment, model, and serial No.
- (3) Specification of the standard reference used to UE the cylinder. Standard reference (calibration ring) must be identified by serial number or other stamped identification marking.
- (4) Cylinder serial number, type, and stamped pressure.
- (5) Maximum allowable filling pressure.
- (6) Minimum prescribed sidewall.
- (7) Number of events at each location
- (8) Description of each AE event (amplitude, duration, energy, etc.).
- (9) Size of each defect measured (length and depth).
- (10) Type of each defect measured (crack, pitting, etc.).
- (11) Defect location relative to each sensor.
- (12) Defect angular location defined by clock direction (3, 5, or 9 O'clock).
- (13) Defect location relative to sidewall (interior, outer surface, inner surface).
- (14) AE and UE technician's name and certification level.
- (15) Test Date.
- (16) Thread inspection results (passed/failed/NA).
- (17) Internal inspection results (passed/failed/NA).

(18) Acceptance/rejection results.

e. PERSONNEL QUALIFICATION: Each person who performs retesting or who evaluates or certifies retest results must meet the following requirements:

(1) Project Manager is the senior manager of Weldship LLC responsible for compliance with DOT regulations including this special permit. Additionally, the project manager must ensure that each operator and senior review technologist maintain the required certifications described herein.

(2) The personnel responsible for performing cylinder retesting under this special permit shall be qualified to an appropriate Level (Level I, II, or III) acoustic emission and ultrasonic examination (AE/UE) in accordance with the American Society for Nondestructive Testing (ASNT) Recommended Practice SNT-TC-1A-1996 depending upon the assigned responsibility as described below:

(i) As a minimum, a Level II Operator must perform system startup, calibrate the system and review and certify the test results when written acceptance and rejection criteria for cylinders has been provided by a Senior Review Technologist. Based upon written criteria, the Level II Operator may authorize cylinders that pass the retest to be marked in accordance with paragraph 7.c. of this special permit. However, a person with Level I certification may perform a system startup, check calibration and perform AE/UE under the direct guidance and supervision of a Senior Review Technologist or a Level II Operator, either of whom must be physically present at the test site so as to be able to observe examination conducted under this special permit.

(ii) Senior Review Technologist (SRT) is a person who provides written AE/UE procedure, supervisory training and examinations (level I and II), technical guidance to operators and reviews and verifies the retest results. The SRT must prepare the reports required in paragraph 7.d. An SRT must have a thorough understanding of the DOT Regulations (HMR) pertaining to the requalification and reuse of DOT cylinders authorized under this special permit and ASNT Recommended Practice SNT-TC-1A and possess:

(A) A Level III certification from ASNT in Acoustic Emission and Ultrasonic Testing;

(B) A Professional Engineer (PE) License with a documented experience for a minimum of 2 years of experience in Non-

Destructive Evaluation (NDE) of pressure vessels or pipelines using the Acoustic Emission and ultrasonic examination techniques; or

(C) A PhD degree in Engineering/Physics with documented evidence of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines using the ultrasonic examination technique or research/thesis work and authoring/co-authoring of technical papers published in recognized technical journals in the fields of Acoustic Emission and ultrasonic testing methods.

The most recent copies of certification (e.g., ASNT Level III or P.E. license) must be available for inspection at each requalification facility.

f. OPERATIONAL CONTROLS:

(1) The maximum fill pressure for any cylinder may not exceed 7/10 of the marked test pressure for that cylinder. The maximum fill pressure for any cylinder covered by this special permit may not exceed 4,000 psi.

(2) No person may perform inspection and testing of cylinders subject to this special permit unless:

(i) That person is an employee of Weldship LLC and has a current copy of this special permit at the location of such inspection and testing; and

(ii) Complies with all the terms and conditions of this special permit.

(3) The marking of the retester's symbol along with the letters AE on the cylinders (when individually tested) and/or affixing a data plate certifies compliance with all of the terms and conditions of this special permit. The data plate must at minimum include the following:

(i) Retesting company name.

(ii) RIN.

(iii) DOT Special permit number.

(iv) Trailer/Module's identification.

(v) Filling pressure (lowest stamped working pressure) at 70 °F.

(vi) Maximum allowed filling pressure including overfill if applicable at 70 °F.

(viii) Test date.

(ix) Retest date.

(x) Water capacity.

(4) Each facility approved by OHMS to test cylinders under the terms of this special permit must have a resident operator with at least a Level II Certification in AE/UE.

(5) In the event of retest with more than one method of retest being performed at the time of retest, the markings per paragraph 7.c. of this special permit shall be displayed only if the final method of retest is as per this special permit.

(6) Cylinders for use in oxygen service must be internally visually inspected and cleaned in accordance with Compressed Gas Association Pamphlet G-4.1, Cleaning Equipment for Oxygen Service, 2004 Fifth Edition.

8. SPECIAL PROVISIONS:

a. The ultrasonic examination (UE) data, results and additional technical information deemed pertinent in successful application of the UE must be recorded and kept at each facility for a minimum of 5 years after completion of UE. For any rejected cylinder, the defect causing the rejection must be fully characterized and profiled. That is, the specific type of defect should be identified (i.e., isolated pits, line corrosion or SBT crack) and the specific size of the defect should be determined (i.e., length, depth, width, diameter, area, etc.). The record includes cylinder type, size, minimum design wall thickness, age, etc. of the rejected cylinder.

b. Shippers (offerors) may use the cylinders specified and tested in accordance with the provisions of this special permit for the transportation in commerce of those hazardous materials specified herein, provided no modifications or changes are made to the cylinders. All terms of this special permit and other applicable requirements contained in 49 CFR Parts 100-185 must be met.

c. Weldship LLC must maintain statements of qualification and supporting information for each “qualified AE/UE tester” who makes use of this special permit. The location of this statement, for each “qualified AE/UE tester,” must be identified to OHMS.

- d. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modifications or changes are made to the package and it is reoffered for transportation in conformance with this special permit and the HMR.
 - e. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, and cargo vessel.
10. MODAL REQUIREMENTS: A copy of this special permit, the revision per which the current retest was performed, must be carried aboard each cargo vessel or motor vehicle used to transport packages covered by this special permit.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR § 171.15 Immediate notice of certain hazardous materials incidents, and

171.16 Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: ae